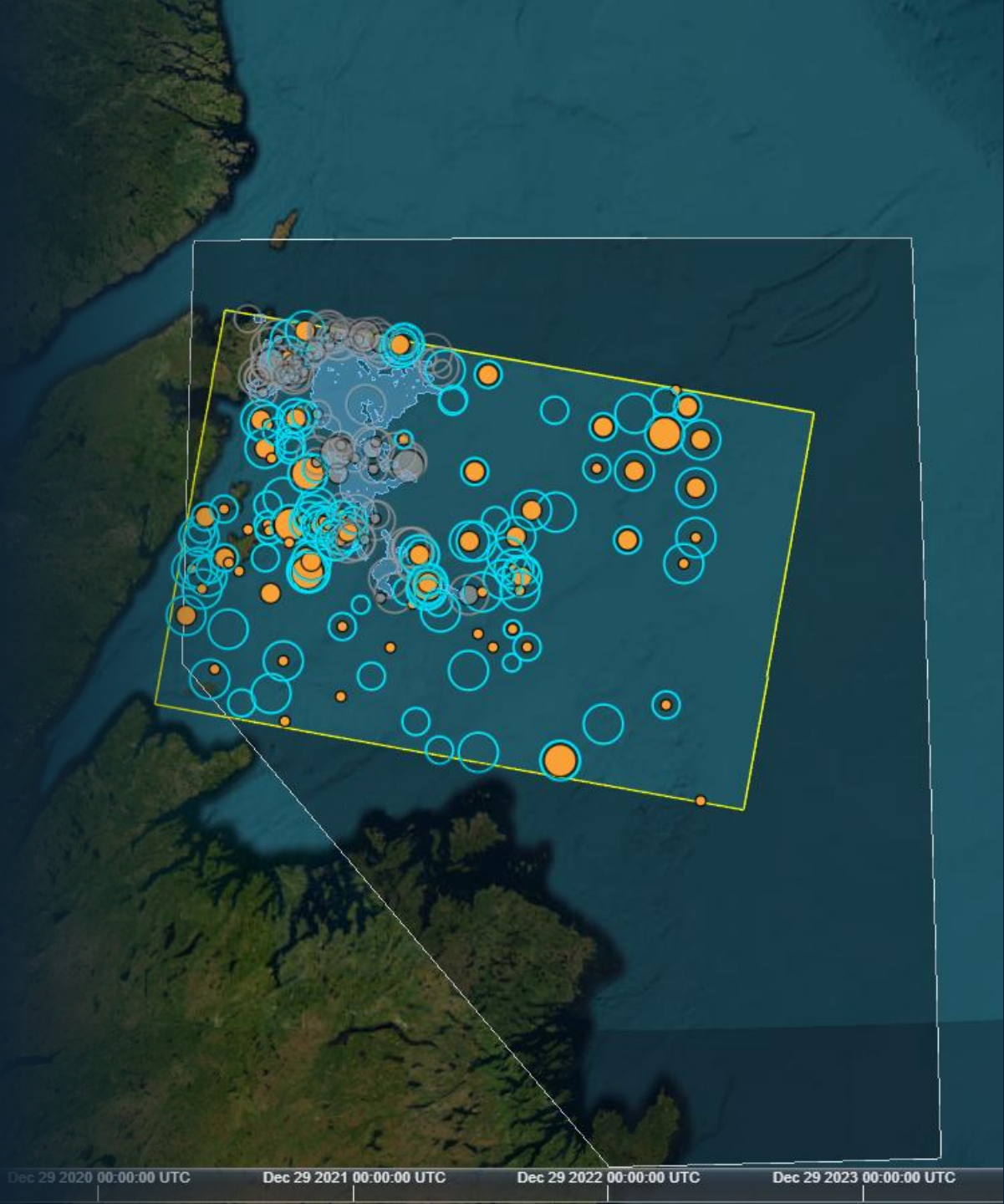
BRIEF · ICEBERG DETECTION

CA-CFAR vs. THOR

A classical radar detector and a SAR foundation model, tested on the same scenes against the same Ice Patrol labels.

Natali Kovalski · October 2026



THE CONTENDERS

Two ways to find a bright berg

CA-CFAR

- **No training:** each pixel vs. its local background, on HV
- **Threshold** from a false-alarm rate and the scene's speckle
- **Filters:** size, ice strips up to 300 m, HH co-pol check
- **CPU:** a full 8-PFA sweep takes about 6 min per scene

THOR + linear head

- **Frozen encoder:** THOR Tiny (FM4CS), pretrained on Sentinel-1
- **769-parameter head** trained on IIP labels, 2019–2020
- **Input:** HH/HV sigma0 in dB, 80 m patches
- **GPU:** 8–15 min per scene on a laptop RTX 3070

TEST SETUP

Everything identical except the detector

Same frames

Four April 2019 passes with same-pass IIP analyst labels: 512 bergs. None was used in training.

Same masks

Land + 500 m, pack ice + 1 km. Detections within 5 km of pack ice are dropped for both.

Same matching

One-to-one within 500 m, with IIP positions moved to their 0.1′ cell centres.

Same budget

Precision needs AIS, so recall is compared at a matched number of detections per frame.

HEADLINE RESULT

+5 pts

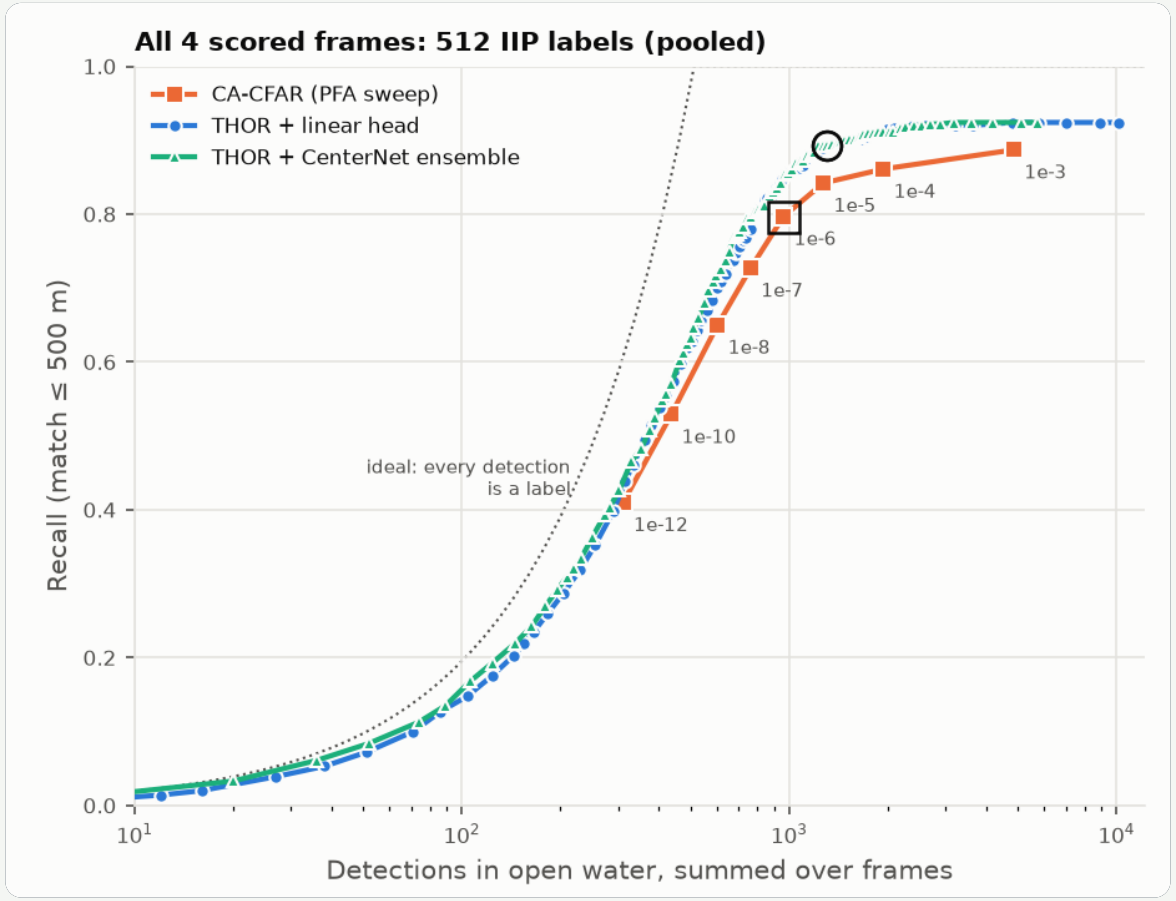
THOR finds 85% of the labelled bergs where the CA-CFAR finds 80%, at the same number of detections: **435 vs. 408 of 512**.

4 held-out frames · 961 open-water detections each · THOR leads on every frame

RECALL VS. DETECTIONS

THOR climbs, CFAR saturates

- **Same recall** as the CFAR with ~18% fewer detections
- **CFAR at its loosest** (PFA 10^{-3}): 0.89 with 4,856 detections
- **THOR** reaches that with ~1,300, and tops out at 0.92
- **Its top ~100 peaks** are almost all labelled bergs



Pooled, 512 IIP labels. Square: CFAR at PFA 10^{-6}. Circle: THOR at score 0.35.

PER FRAME

Ahead on every frame, by 2 to 7 points

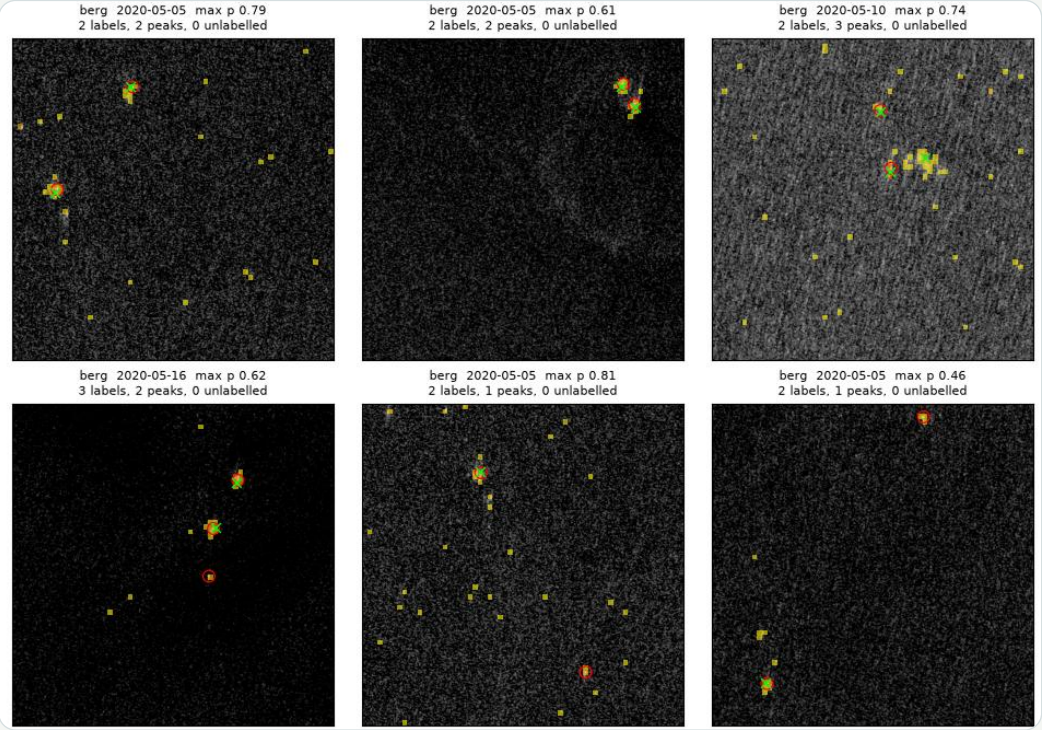
Frame	IIP labels	Detections	CA-CFAR	THOR
2019-04-29	267	508	0.83	0.90
2019-04-23	150	278	0.79	0.82
2019-04-24	49	47	0.82	0.84
2019-04-17	46	128	0.61	0.67
Pooled	512	961	0.80	0.85

04-17 is low for both: 7 of its 46 bergs lie within 5 km of pack ice, where both detectors are filtered out. The CFAR's strip rule is off on 04-29 and 04-24 (no pack ice).

WHERE THE GAIN IS

Medium bergs, not small ones

Bergs (pooled)	CFAR	THOR
Small, 15–60 m (73)	0.71	0.71
Medium, 61–120 m (425)	0.81	0.88
Large, > 120 m (14)	10	11
> 20 km from ice (275)	0.88	0.94
Incidence 40–47° (398)	0.80	0.84



THOR heatmaps on 2020 validation chips: IIP labels (red circles), THOR peaks (green ×). The CFAR's 300 m strip rule drops large bergs; it now runs only in scenes with pack ice.

LIMITS

THOR doesn't replace the pipeline

01

Pack ice

Alone, it fires on pack ice: 34–82% of its peaks on icy frames fall inside the CFAR's ice mask.

02

Small bergs

A tie at 0.71. An 80 m patch is coarse for a 20 m target; 4×4 patches are the next test.

03

Precision

Unknown for both without AIS. At the matched count, CFAR has 553 and THOR 526 unlabelled detections.

VERDICT

THOR on top of the CFAR pipeline

What holds

- **+5 points recall** at an equal number of detections
- **Large bergs:** 11 of 14 found, CFAR 10
- **Needs the pipeline:** its pack-ice mask, and a GPU
- **Caveat:** one month of one year; THOR learned the analysts' labels

Next

- **AIS for the test frames** to measure precision for both
- **Shape-based strip test** so the CFAR keeps large bergs near ice
- **THOR at 4×4 patches** for small bergs
- **More years** and near-range frames

THANK YOU

Questions?

Natali Kovalski · github.com/natali-kovalski/iceberg-alley-sar

Live viewer with both detectors: natali-kovalski.github.io/iceberg-alley-sar. Write-up: docs/thor-vs-cfar.md. Contains modified Copernicus Sentinel data (2019–2026). Labels: International Ice Patrol (NSIDC G00807). THOR: FM4CS (Apache 2.0); Forgaard et al., IGARSS 2026.